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Characterization and Inverse Design of Stochastic Mechanical Metamaterials Using Neural Operators

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Machine learning (ML) is emerging as a transformative tool for the design of mechanical metamaterials, offering properties that far surpass those achievable through lab-based trial-and-error methods. However, a major challenge in current inverse design strategies is their reliance on extensive computational and/or experimental datasets, which becomes particularly problematic for designing micro-scale stochastic architected materials that exhibit nonlinear mechanical behaviors. Here, a comprehensive end-to-end scientific ML framework, leveraging deep neural operators (including DeepONet and its variants) is introduced, to directly learn the relationship between the complete microstructure and mechanical response of architected metamaterials from sparse but high-quality in situ experimental data. Various neural operators and standard neural networks are systematically compared to identify the model that offers better interpretability and accuracy. The approach facilitates the efficient inverse design of structures tailored to specific nonlinear mechanical behaviors. Results obtained from stochastic spinodal microstructures, printed using two-photon lithography, reveal that the prediction error for mechanical responses is within a range of 5 - 10%. This work underscores that by employing neural operators with advanced nano- and micro-mechanical experiments, the design of complex micro-architected materials with desired properties becomes feasible, even in scenarios constrained by data scarcity. This work marks a significant advancement in the field of materials-by-design, potentially heralding a new era in the discovery and development of next-generation metamaterials with unparalleled mechanical characteristics derived directly from experimental insights.

1. Introduction

Designing material architectures at different scales, including truss-, plate-, or shell-based microstructures, unlocks unique properties distinct from their bulk counterparts, thereby significantly expanding the current material design space.[1–3] For instance, periodic micro- and nano-lattices, fabricated from a wide variety of materials, exhibit extraordinary mechanical properties such as ultra-high specific strength,[4,5] energy absorption,[6,7] and impact resilience.[8] Likewise, origami metamaterials exhibit auxeticity, tunable mechanical properties, and recoverability.[9,10] These architected materials offer unprecedented flexibility in the design of structures with desired properties, which paves the way for designing the next-generation engineering materials.[11] Traditionally, such a design process has been intuition-based, often inspired by natural cellular materials, such as bones or nacres.[12–14] More generally, the conceptualization of metamaterials with non-periodic microstructures, e.g., spinodal topology, exhibits high mechanical resilience and tunable mechanical anisotropy.[15,16] These microstructures

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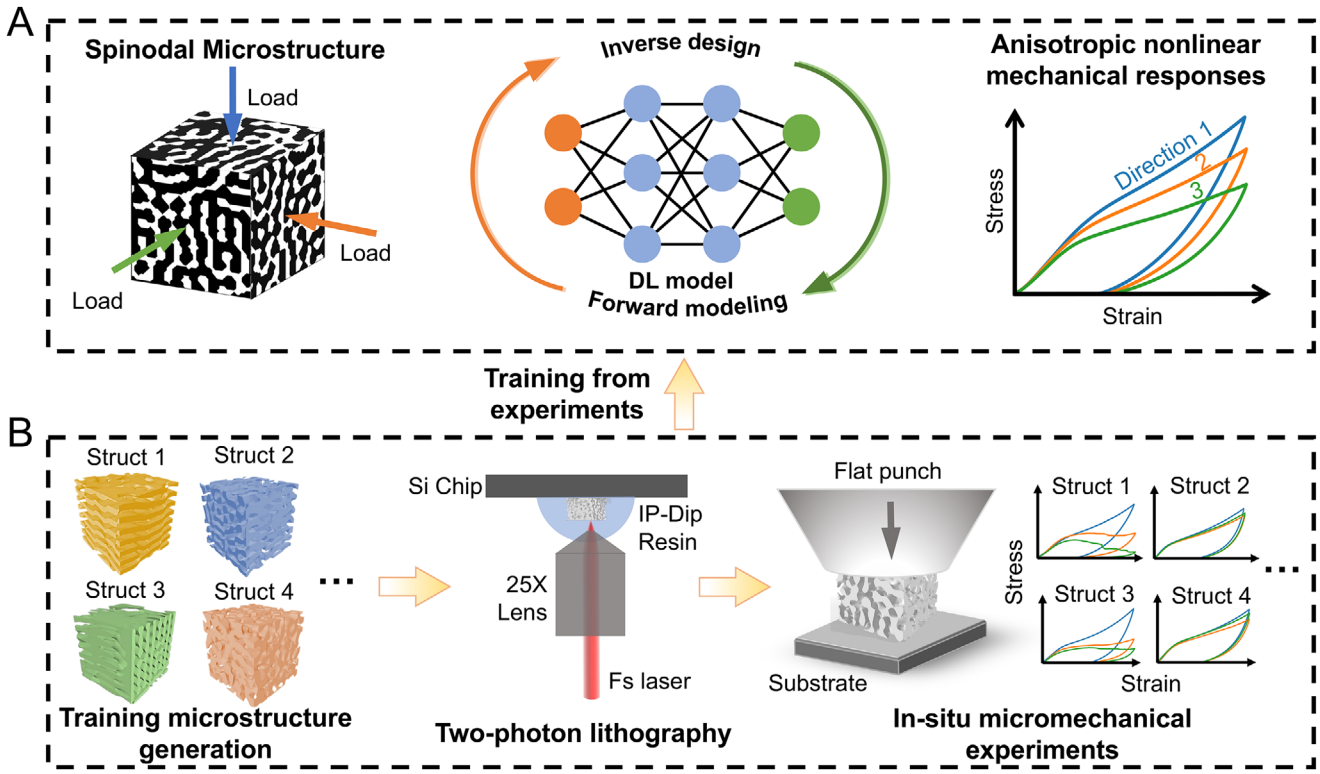


Figure 1. Forward Modeling and Inverse Design of Anisotropic Stochastic Micro-architected structures using Deep Learning based on Limited and Sparse in situ Micro-Mechanical Experimental Data. A) Schematic of forward modeling and inverse design; B) Training data collection from in situ micro-compression experiments on samples fabricated by two-photon lithography.

were inspired by bi-continuous phases found in bones and the spinodal decomposition observed during copolymer synthesis.[17,18] However, such an intuition-based design approach is limited by the random exploration of the design space. To overcome these limitations, there is a growing need for inverse design methodologies, which enable the purposeful creation of architectures tailored for specific mechanical properties, rather than relying on serendipitous discovery.

In this context, the emergence of machine learning (ML) tools has been promising, offering innovative approaches to derive insights from both experimental and computational data, thereby paving the way for inverse design.[19–25] Facilitated by ML, the design process is gradually steering from a heuristic process towards a more systematic and data-driven approach. However, this pivotal transition is contingent upon the availability of sufficient training datasets, typically sourced from computations such as finite element analysis (FEA) and large-scale molecular dynamics (MD). Moreover, the approach poses significant challenges when applied to complex material architectures due to the computational complexity of non-periodic structures and their multiscale material/structural responses, including nonlinearity, instabilities, and damage. Indeed, the inherent complexity of stochastic architected materials and their associated responses results in a scarcity of robust and reliable computational training datasets for ML applications, thereby creating a bottleneck for progress. Furthermore, defects and imperfections created during the fabrication process and the dependency of mechanical properties on process parameters mean that ML predictions can notably deviate from actual experimental outcomes.[26,27] Therefore, the inverse design of these materials has been limited primarily to simple design criteria such as homogenized elastic moduli.[22,28,29] Recently, an approach for the inverse design of metamaterials beyond linear elasticity was advanced, which involves training an ML model directly from experimental datasets.[24] While compelling, this approach has the potential drawback that it requires copious training datasets, with the associated fabrication and characterization burden. While some examples of autonomous experimentation have been reported for designing macro-scale metamaterials,[30,31] the field is still in its infancy, especially for conducting sophisticated in situ experiments needed for nano- and micro-architected material design. Therefore, novel approaches for inverse design of multifunctional materials with complex architectures and desirable properties beyond elasticity, e.g., nonlinear responses associated with material nonlinearities, instabilities, and damage, are highly needed.

Herein, we address these challenges by introducing an innovative, end-to-end scientific machine learning (SciML) framework[32] based on the deep neural operator (DeepONet),[33] which can effectively design metamaterials based on sparse and limited experimental data (**Figure 1A**). Despite our framework being scale-agnostic, enabling the design of architected materials across multiple scales, we focus on the inverse design of micro-architected spinodal metamaterials as a proof of concept, targeting desired mechanical responses beyond linear elasticity. We collected a sparse yet high-quality experimental dataset from in situ micro-compression experiments conducted inside a

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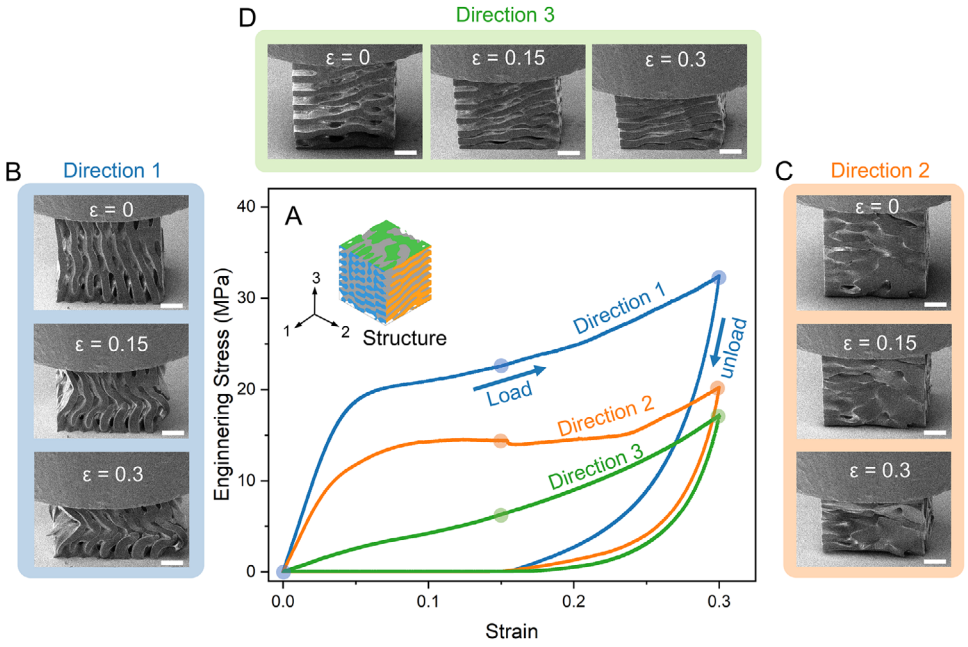


Figure 2. In situ Micro-Compression Experiment for a Spinodal Microstructure. A) Nonlinear and anisotropic stress-strain curves of compression test along three principal directions. Insets show the microstructure design and Cartesian coordinates; B–D) SEM images of the samples deformed in direction 1, 2, 3, respectively, at three different strain levels. The scale bar is 20 μm.

scanning electron microscope (SEM) on structures fabricated through two-photon lithography (TPL) (Figure 1B). Our results demonstrate that the enhanced DeepONet is a promising surrogate model to predict the mechanical responses of micro-architected materials with a prediction error of 5 – 10%. We also introduce a successful implementation of inverse design for desired nonlinear stress–strain behaviors while avoiding unreliable and expensive computations. We anticipate that our methodology will lay the foundation for developing next-generation materials with tailored properties and functionalities when only limited data is available.

2. Results

2.1. Mechanical Data Collection from in situ Micro-Mechanical Experiments

The mechanical behavior of spinodal metamaterials, such as stiffness and energy absorption, can be precisely adjusted through their microstructural design.[15] These structures often exhibit marked anisotropy in their mechanical responses. **Figure** 2A illustrates this phenomenon with a lamellar-type microstructure fabricated using TPL, showcasing pronounced mechanical anisotropy. When the microstructure was loaded in direction three, strain-hardening was evident. We attribute such behavior to the densification of the lamellar microstructure (Figure 2D; Movie S3, Supporting Information). In contrast, loading in directions one and two displayed negligible strain-hardening owing to structural buckling and crack formation (Figure 2B, C; Movies S1 and S2, Supporting Information).

It is also worth noting that the mechanical properties can be largely influenced by the TPL fabrication parameters, such as laser power, slice/hatch distances, and the choice of photoresists, as well as the testing environment. Consequently, our objective is to predict and design microstructures that exhibit specific mechanical behaviors while maintaining a consistent fabrication process and using the same base material, all under consistent environmental conditions inside the SEM chamber (vacuum and room temperature). To this end, we performed in situ micro-compression experiments on 29 representative micro-scale spinodal structures fabricated via TPL. To account for both fabrication and structural uncertainties in the DeepONet training, we conducted repeated tests across seven different microstructures using the same and different random seeds (See Supporting Information). The complete experimental datasets are shown in Figure S1 (Supporting Information). These tests provided a sparse yet high-quality dataset of stress–strain curves for training the DeepONet. The experimental procedures are presented in Experimental Section. Details regarding the generation of stochastic spinodal microstructures are presented in Supporting Information.

2.2. Two-Step DeepONet (QR) for Relating Microstructures with Nonlinear Mechanical Properties

In this study, we employed DeepONet and its variants[33,34] to develop a SciML model tailored for analyzing the mechanical properties of materials with spinodal microstructures. DeepONet is an innovative architecture of neural networks specifically designed to learn (nonlinear) operators, i.e., mappings from functions to functions.[35] Since its invention,[33] DeepONet has been successfully applied to develop surrogate models for various engineering problems, including multiscale mechanics,[35] phase

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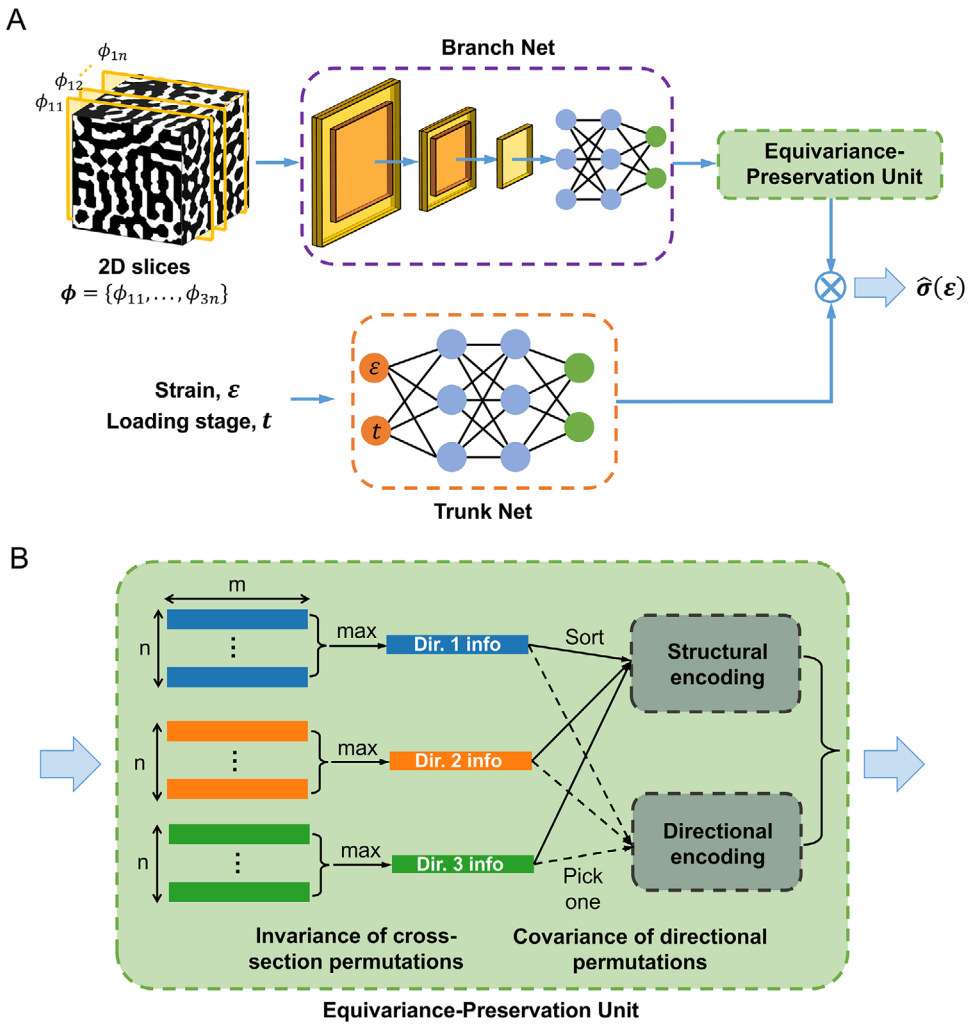


Figure 3. DeepONet Structure for Relating Microstructures with Stress-strain Curves of Spinodal Materials. A) Schematic of the DeepONet architecture. The microstructure is preprocessed into 2D cross-sections, denoted as $\{\phi^{(rs)}\}_{r,s=1,1}^{3,n}$. These cross-sections are sequentially analyzed by the branch Net and the equivariance-preserving unit. The loading strain ϵ and the pseudo-time t (indicating the loading/unloading step) are fed into the trunk Net. The final output is the predicted stress $\hat{\sigma}$ corresponding to the input microstructure for the given strain ϵ and loading step t. B) Detailed structure of the equivariance-preserving unit; m represents the dimension of the output vector from the branch Net for each cross-section. This unit is designed to maintain invariance to permutations of cross-sections and covariance to directional permutations. This is achieved through max pooling for cross-section permutations (on the left) and structural and directional encoding for directional permutations (on the right).

transitions,[36] ocean engineering,[37] and biomechanics.[38,39] The DeepONet architecture comprises a branch Net and a trunk Net, as illustrated in **Figure** 3A. Briefly, it maps the microstructural information into the corresponding stress–strain curves for a complete loading-unloading cycle. Recent studies on the training and generalization of DeepONets have demonstrated that a two-step training method,[34,40] where the trunk Net is trained first, followed by training the branch Net, yields better generalization compared to training both networks simultaneously. In this work, we predominantly employ the two-step DeepONet (QR),[34] and compare the accuracy and efficiency with standard neural networks, such as UNet,[41] and other neural operators like Fourier Neural Operator (FNO).[42] The detailed discussions on interpretability of two-step DeepONet can be found in Supporting Information. We briefly summarize the two-step DeepONet (QR) architecture in this section and provide complete technical details in Experimental Section and Supporting Information.

The two-step DeepONet (QR) consists of a branch Net and a trunk Net,[34] and it employs a two-step training strategy, as illustrated in Figure S4 (Supporting Information). In the first training step, Figure S4A (Supporting Information), the trunk Net is trained along with a randomly initialized matrix A. Upon completion of trunk Net training, a QR decomposition is performed for the trunk Net T to obtain R^*, where $Q^*R^* = qr(T)$. In the second step, the branch Net is trained to approximate $R^*\hat{A}$, where $\hat{A}$ is the optimized A in the first step. Finally, the branch Net is multiplied with the trunk Net, which has already been multiplied by R^{*-1}, as shown in Figure S4B (Supporting Information). Moreover, we modify the basic architecture to endow it with an additional equivariance-preserving unit that accommodates the

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inherent symmetries specific to our study. The branch Net takes the individual 2D cross-sections of the 3D microstructures as inputs, represented as $\{\phi^{(rs)}\}_{r,s=1,1}^{3,n}$, where r indicates the normal direction, s the index of the cross-section, and n the total number of cross-sections. The trunk Net inputs include the loading strain ϵ and the pseudo-time t, the latter of which indicates whether the material is in the loading or unloading step. An equivariance-preserving unit is appended after the branch Net. The final output of two-step DeepONet (QR) is the stress prediction $\hat{\sigma}$ corresponding to the input microstructure $\{\phi^{(rs)}\}_{r,s=1,1}^{3,n}$, evaluated at the particular strain ϵ and loading stage t. This stress prediction is derived by performing a "dot product" of the output vectors from the equivariance-preserving unit and the trunk Net.

The equivariance-preserving unit, illustrated in Figure 3B, is designed to incorporate the inherent symmetries into the two-step DeepONet (QR) branch Net. The input of this unit consists of $3n$ vectors, each of dimension m, originating from the branch Net. We consider two types of symmetries, namely, the invariance of cross-section permutations and the covariance of directional permutations. The invariance of cross-section permutations means that permuting the sequence of the n cross-sections within the same normal direction should not affect the resulting output stress–strain curve. While this does not hold precisely, it is a practical simplification adopted in our study to reduce the complexity and dimensionality of the input space. Although allowing the shuffling of these $x-y$ cross-sections, for example, disrupts the original sequential order of the cross-sections along the z direction, we rely on $x-z$ and $y-z$ cross-sections to retrieve the sequential information along z direction. The covariance of directional permutations, on the other hand, implies that permuting the three spatial directions of the microstructure results in a corresponding permutation at the output of the equivariance-preserving unit. The implementation of these two types of symmetries is achieved technically through max pooling and sorting, respectively, both of which have been applied in existing literature on neural networks.[43,44] Max pooling (see the left side of Figure 3B) is applied on the n vectors outputted from individual cross-sections along the same direction, yielding direction-wise information vectors. The sorting operation (see the right side of Figure 3B) is applied on the three direction-wise vectors in an element-wise way, outputting a so-called structural encoding. Meanwhile, one of the three direction-wise vectors is picked as the directional encoding to evaluate the mechanical property for a certain spatial direction. The combination of two encodings serves as the output of the equivariance-preserving unit. Further details and discussions on this equivariance-preserving unit can be found in the Supporting Information. To train and evaluate the modified two-step DeepONet (QR), we used the experimental stress–strain data as well as the augmented microstructure dataset. The augmented microstructure dataset comprises the microstructures actually printed experimentally as well as those variants with similar patterns that are generated with the same set of phase angle parameters (see Supporting Information). We assumed that microstructures created with identical phase angle parameters would exhibit the same mechanical behaviors. We trained the two-step DeepONet (QR) using the experimental data for a total of 16,000 epochs - 80,000 epochs for each training step. We observed that both the training and testing losses decayed rapidly, as depicted in Figure S5 (Supporting Information), which indicated that two-step DeepONet (QR) can well-capture the relationship between the spinodal microstructure and the nonlinear stress–strain curves during both loading and unloading.

2.3. Predicting Mechanical Responses of Unseen Micro-architectures

We first evaluate the performance of the trained two-step DeepONet (QR) in predicting the mechanical properties of previously unseen microstructures. The predicted stress–strain curves along the three principal directions are presented in **Figure 4A**. In each case, we compare the experimental data (dashed lines) with the predictions from the two-step DeepONet (QR) and the FNO (solid lines). The DeepONet demonstrates remarkable accuracy in predicting both the loading and unloading stress–strain curves in all three directions, achieving the mean squared error (MSE) for all testing cases of 4.69×10^{-3}. This high level of precision is particularly noteworthy given the limited experimental data available for training. Notably, the DeepONet not only accurately predicted strain hardening but also adeptly captured strain-softening behaviors resulting from structural buckling and failure, demonstrating its capability of handling highly nonlinear material behavior. Notable examples include the prediction for direction 1 of the (0, 15, 15) sample and directions 1 & 2 of the (0, 0, 30) sample. While the FNO captures the overall trends of the stress–strain curves, its accuracy is significantly lower compared to the prediction made by the two-step DeepONet (QR). This discrepancy may be attributed to that the two-step DeepONet utilizes two networks, allowing it to take both microstructure and strain as inputs, whereas the FNO relies solely on microstructures inputs.

To quantitatively evaluate the prediction performance of the two-step DeepONet (QR), we present a detailed comparison of predicted versus experimental values for all stresses, energy absorption, ultimate stress, and stiffness across all test cases in Figure 4B. In each subplot, the red axes represent the experimental values compared with those predicted by the two-step DeepONet (QR), while the blue axes compare the experimental values with predictions made by the FNO. The predictions of the two-step DeepONet (QR) for all stresses, energy absorption, ultimate stress, and stiffness show high accuracy, with a coefficient of determination (R^2) over 0.98, which are higher than that predicted by the FNO. Especially, the R^2 value of the FNO for Energy Absorption is 0.836, which is much lower than that of the two-step DeepONet (QR) (0.985). Two-step DeepONet (QR) demonstrates remarkable efficacy in predicting the mechanical properties of unseen microstructures, highlighting its potential as a robust tool for analyzing mechanical behaviors in various architectures, especially in situations where experimental data availability is limited.

2.4. Inverse Design and Experimental Validation

We have successfully demonstrated the efficacy and accuracy of the DeepONet model, primarily as a surrogate for forward modeling. However, the capabilities of DeepONet extend well beyond

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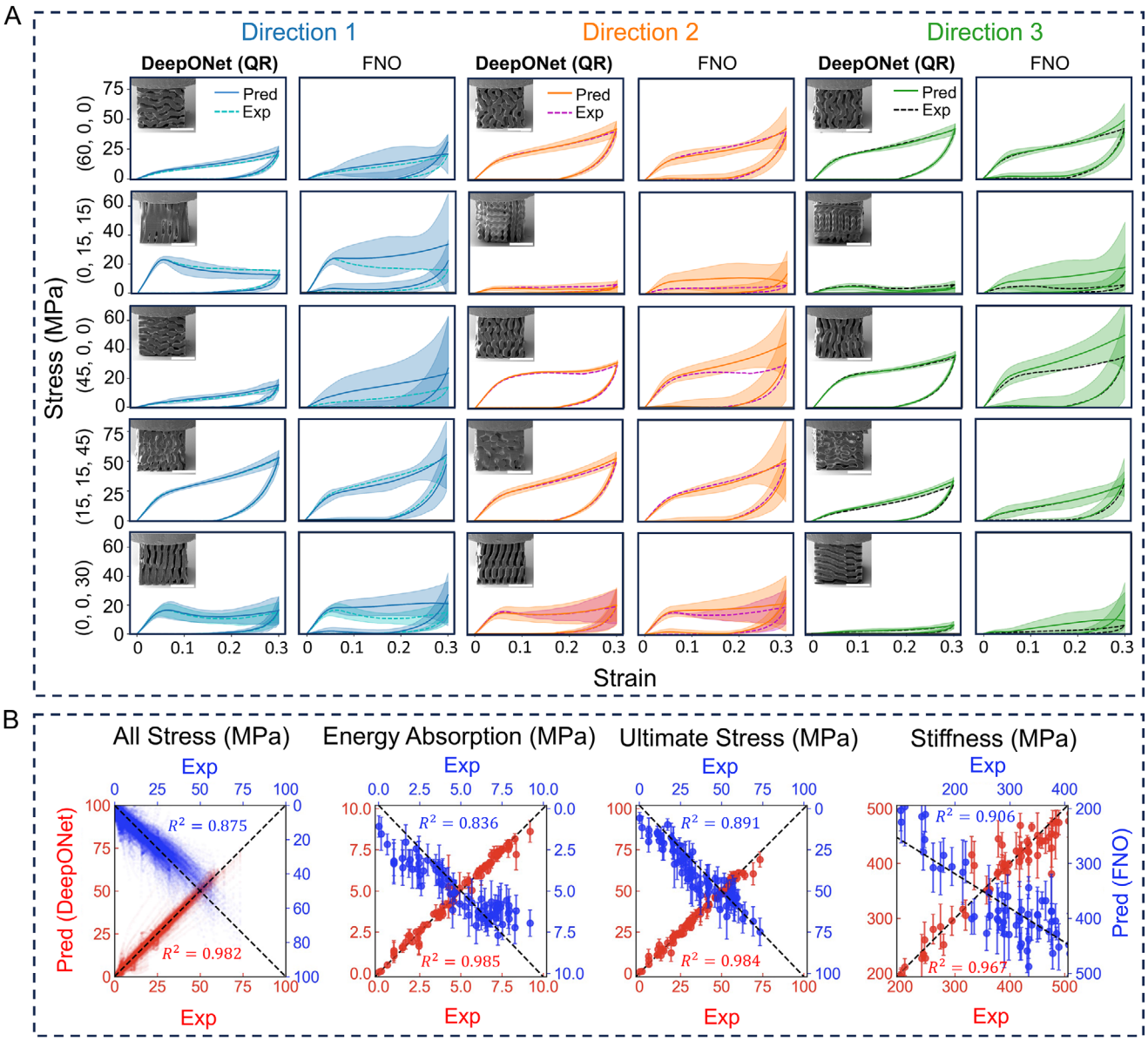


Figure 4. Performance of Two-step DeepONet (QR) and Fourier Neural Operator (FNO) in Predicting Mechanical Properties of Unseen Microstructures. A) The experimental data and the predicted stress–strain curves of five typical spinodal microstructures based on Two-step DeepONet (QR) and the FNO. The dashed lines represent the experimental data, and the solid lines represent the predictions. The shaded area is two times of standard deviation. Insets show the corresponding spinodal microstructure. The corresponding phase angles are in the left of their respective panels. The scale bar is 50 μm. B) Comparison of predictions and experimental values of stress values, energy absorption, ultimate stress, and stiffness. The red axes are the experimental values and values predicted by DeepONet, and the blue axes are the experimental values and values predicted by the FNO.

the forward modeling. In this section, we delve into the potential of our method for the inverse design of microstructures tailored to specific mechanical properties. While a comprehensive inverse design process involves forward simulation/modeling, optimization algorithms, generation of candidate designs, defining performance criteria and objectives, and iterative feedback, we herein utilized a streamlined approach consisting of an indirect inverse design methodology. This approach does not rely on direct generative modeling or gradient-based optimization but rather leverages DeepONet as a fast and accurate structure-property predictor to rapidly identify the best design. The framework of the inverse design workflow is illustrated in Figure S13 (Supporting Information). This process exclusively involves the forward propagation through two-step DeepONet (QR), enabling its completion within seconds.

We provided four representative stress–strain responses, as targets, which were not included in the DeepONet training data. These target curves were generated by mixing and/or averaging the curves from two experimental samples. After identifying the best design, we compared three sets of stress-strain curves: 1) the target stress–strain curves, 2) the DeepONet-predicted stress–strain curves for the designed microstructures,

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and 3) the experimental stress–strain curves of the designed microstructures.

The outcomes for these four cases are displayed in **Figure 5**, which include the target curves, the chosen designs, the fabricated microstructures, and a comparison of the three sets of curves. In **Case I**, our objective was to design a microstructure with isotropic mechanical responses (Figure 5A). We identified the best design which has a relative $\mathcal{L}_2$ error of 1.18% (averaged in 3 directions) between predicted and target stresses-strain behaviors. Experimental verification revealed an error of 5.29% compared to the target stress–strain behavior. Considering the existence of stochastic fabrication defects and instrumentation errors during testing, the result represents a remarkable accuracy for designing isotropic spinodal microstructures. Movies S4–S6 (Supporting Information) display the in situ SEM experimental verification for **Case I** under three principal directions, respectively. For **Case II**, we intended to design a microstructure with large variation in the mechanical response along all three directions (Figure 5B). The relative $\mathcal{L}_2$ error between the target and predicted stress–strain behaviors was 3.52%, and the relative $\mathcal{L}_2$ error between the experimental and targeted stresses was 15.69%, again demonstrating relative high accuracy. In **Case III**, our goal was to design a microstructure with strong anisotropy with slight strain-softening behaviors in two directions (Figure 5C). The inverse design algorithm selected a lamellar-type microstructure, exhibiting moderate strain softening in directions one and two (due to lamellar buckling and failure), while a strain hardening behavior in direction three (due to densification of the lamellae). The model predictions closely matched the target, with a relative $\mathcal{L}_2$ error of 6.54% between the predicted and target curves. The experimental data showed a larger error of 24.13%, but still fell within the 95% confidence interval, indicating good alignment between the experimental and target curves. This highlights the algorithm's capability to design microstructures with desired anisotropic strain-softening properties with strain-softening behaviors. **Case IV** focused on designing a microstructure exhibiting pronounced strain-softening in two directions due to structural bulking (Figure 5D; Movies S7–S9, Supporting Information). The relative $\mathcal{L}_2$ error between the target and predicted stress–strain behaviors was 14.61%, and the relative $\mathcal{L}_2$ error between the experimental and target curves was 22.96%, still within the 95% confidence interval. Again, the experimental results confirmed the predicted strain-softening behavior. Despite the higher error in direction two, we attribute this to imperfections in compressive buckling behaviors rather than the limited training data obtained from strain-softening cases. This assertion is based on the fact that the DeepONet algorithm is capable of accurately predicting strain-softening behaviors in previously unseen microstructures (Figure 4).

3. Discussion

Collecting high-fidelity experimental data from in situ micro- and nano-scale mechanical tests is a resource-intensive task, both in terms of time and cost. However, this data collection remains the most reliable method for obtaining precise mechanical information at small scales. This necessity underscores the importance of complementary SciML methods that can effectively build the surrogate models, particularly given the challenges posed by data scarcity as well as inherent nonlinearities and softening in mechanical responses.

In this work, we introduced a SciML approach specifically tailored to learn from limited experimental data on the anisotropic, nonlinear mechanical properties of micro-architected spinodal materials. To achieve the desired functionality, we carefully designed the SciML framework and processed the experimental data with several key strategies. First, we chose DeepONet as the base model, known for its efficacy as a nonlinear operator approximator and strong generalization capabilities for unseen testing data.[33] This foundational architecture is crucial for accurately capturing the complex relationship between microstructure and nonlinear mechanical behaviors. Second, we enhanced the DeepONet by integrating symmetries and prior assumptions directly into the network architecture as inductive biases. This integration was achieved through the equivariance-preserving unit mentioned earlier. Third, we reduced the dimensionality of the input data by extracting representative 2D cross-sections from the original 3D structures, as shown in Figure 3A). Additionally, we augmented the dataset by generating similar microstructures using the underlying sampler, while maintaining the assumption that these synthetic microstructures exhibit the same mechanical behavior as their real counterparts. These methodologies collectively empower our SciML approach to achieve robust predictive capabilities, significantly reducing the reliance on extensive training data from raw in situ experiments.

While the study here reported primarily focused on micro-scale spinodal architected metamaterials, it is important to recognize the broader applicability of the method to a wide range of other mechanical metamaterials with diverse mechanical properties. This versatility stems from DeepONet's robust capability for approximating and generalizing nonlinear function-to-function relationships. However, it is important to acknowledge that the presented workflow, particularly in the investigated scenario of limited data availability, may not be universally applicable to all base material types. The photoresist we used in this study, IP-Dip, typically exhibits smooth and reproducible stress–strain curves, as shown in Figure S3 (Supporting Information). These mechanical behaviors are a key factor in the success of the approach relying on very limited experimental data. In contrast, materials with inherently more brittle characteristics, such as glassy carbon,[4,45,46] or those with complex microstructures like nanocrystalline metals,[47] often display much more stochastic mechanical responses due to the random distribution of defects within these materials. For such materials, where mechanical responses are significantly influenced by material stochasticity, advanced SciML methods like statistical learning, Bayesian optimization,[48] and uncertainty quantification[49] can be employed to extend the framework here advanced. These hybrid approaches should be capable of accounting for the inherent randomness in mechanical properties, thereby providing a more accurate model for these types of materials still with limited experimental data. Moreover, when advanced simulation tools, such as reduced-order models for stochastic metamaterials, become available, a multi-fidelity SciML model could be employed.[50–52] Such a model would leverage both cost-effective, low-fidelity simulations, and expensive, high-fidelity experimental data, enabling the inverse design of a +wide range of metamaterials with

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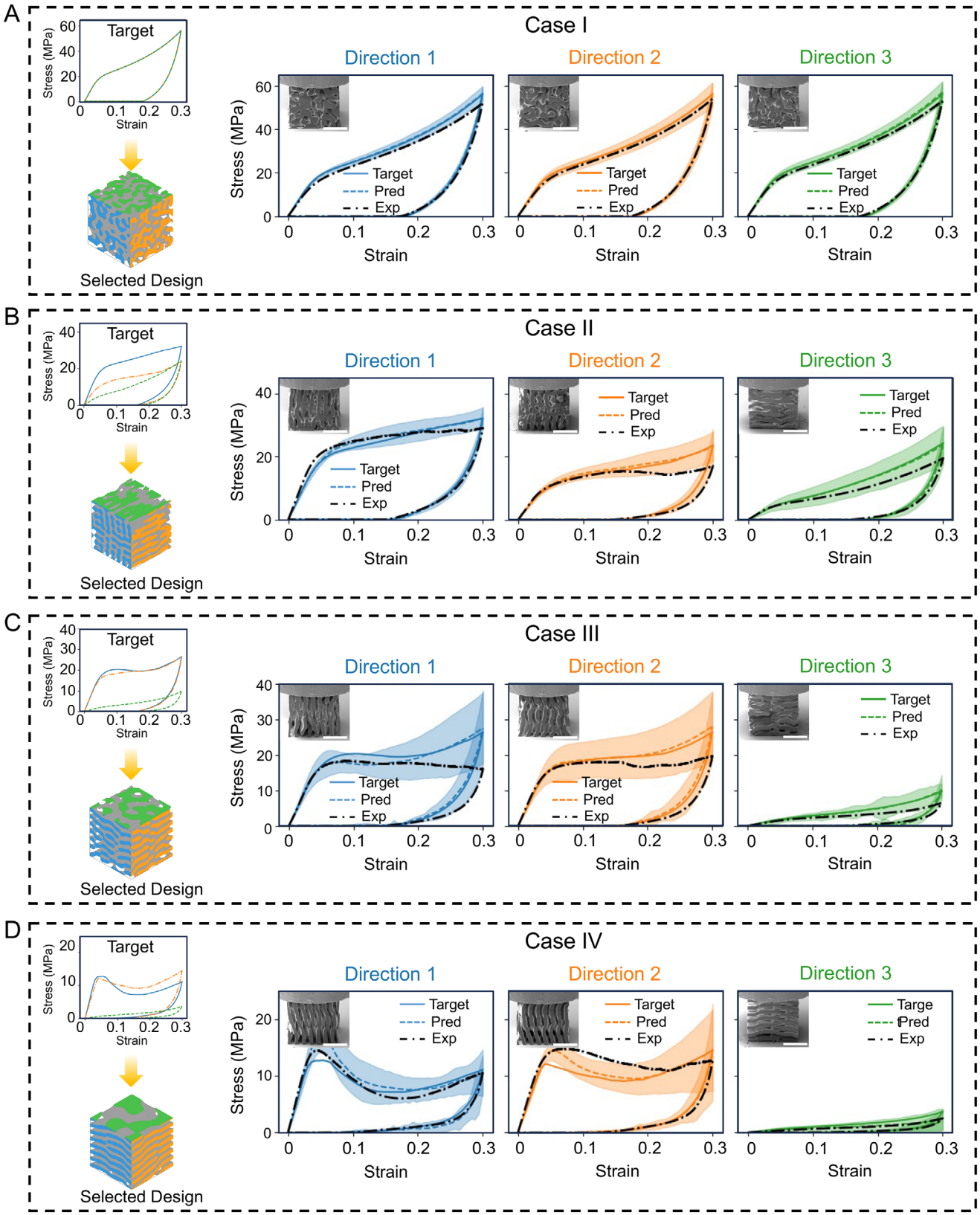


Figure 5. Results of the Trained DeepONet in the Inverse Design of Spinodal Microstructures with Desired Mechanical Responses. A) Case I, B) Case II, C) Case III, and D) Case IV. Displayed curves are the target stress–strain curves (colored solid line) and those predicted by DeepONet (colored dashed line), alongside the experimental data (black dashed line), for comparison in three directions. Insets are SEM images of designed structures. The scale bar is 50 µm.

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complex material and structural behaviors, bridging the gap between simulations and real-world experimental data.

In this study, we have developed a novel SciML model based on DeepONet to model the relationship between microstructure and the nonlinear mechanical properties of micro-architected spinodal metamaterials, utilizing a limited but high-quality experimental dataset from in situ SEM experiments on TPL-fabricated samples. We first showed that the trained DeepONet can accurately predict the nonlinear, anisotropic mechanical properties of various microstructures, including those not included in the training data. Subsequently, we leveraged the trained DeepONet for the inverse design of microstructures, aiming to achieve the desired stress–strain behaviors. Through experimental validation, we showed that the selected design is accurate for target stress–strain curves that are unseen in the training stage. These results demonstrate the effectiveness of integrating state-of-the-art SciML algorithms with advanced nano- and micro-scale fabrication and characterization techniques in designing complex metamaterials, even under data scarcity. We highlight that the designed stochastic metamaterials with tailored mechanical responses can lead to technological advancements, from aerospace engineering to transportation to biomedical devices. As such, the present work offers a new perspective in the field of materials-by-design, potentially paving the way for the discovery of next-generation multi-functional metamaterials that combine mechanical resilience with energy efficiency, durability, and environmental sustainability.

4. Experimental Section

Sample Fabrications: The micro-architected spinodal structures were fabricated through a two-photon lithography system (Nanoscribe, GmbH), using the commercial two-photon photoresist (IP-DIP). The structures were printed on Silicon wafers using the dip-in mode with a 25x objective lens. To enhance the adhesion between the structures and the substrate, the Si wafers were cleaned using isopropyl alcohol (IPA) followed by de-ionized (DI) water. The writing speed and the femtosecond laser power were set as 10 μms^{-1} and 34 mW to yield optimal printing quality. The microstructures were printed in an array with spacing of 450 µm. A 10-min pause was introduced between the printing of each structure to mitigate the impact of laser-induced temperature increases, ensuring consistent mechanical properties across all printed structures. After printing, the structures were developed in propyleneglycol-monomethyl-ether-acetate (PGMEA) for 20 min and thereafter immersed in IPA for 15 min to remove residual PGMEA. Finally, the structures were dried using a critical point dryer (Tousimis, Automegasamdri-915B, Series B) to avoid collapse or distortion that could occur due to surface tension during solvent evaporation. For each microstructure, printing was carried out along three principal directions for micro-mechanical testing. For the anisotropic microstructures, one or two directions exhibited horizontal lamellar structures. These horizontal overhang layers lacked sufficient support along the printing direction, potentially leading to inaccuracies during printing and significantly affecting the mechanical testing results. To address this, the microstructures with horizontal overhangs were printed in the other orientation and then carefully flipped to the target direction using a micromanipulator (The Micromanipulator Company, Model 550) on a probe station (The Micromanipulator Company, Model 6000). The silicon wafer containing the microstructures was then glued to an SEM pin stub (Ted Pella) using super glue (Loctite 460). Then, the samples were uniformly coated with 8 nm Osmium using an Osmium Plasma Coater (OPC-60A, SPI) to reduce charging during SEM observation, which had negligible effects on their mechanical behaviors.

In Situ Micromechanics Experiments: Uniaxial compression experiments were conducted in situ SEM (Thermo Fisher, Scios 2), using a commercially available nanomechanical testing platform (Alemnis AG) with displacement control capability (Figure S2, Supporting Information). The displacement field was prescribed by a programmable piezo actuator paired with a 200 µm diameter flat punch. The resolution of the piezo actuator was 1 nm. The punch speed was set to maintain a quasi-static strain rate of $2 \times 10^{-3} s^{-1}$. The load was gauged using a load cell with a capacity of 1 N and 4 µN resolution. The displacement profile was corrected for the compliance of the instrumentation and substrate. Subsequently, the engineering stress–strain profiles were obtained from corrected load-displacement profiles by normalizing the samples' footprint area and height, respectively. The energy absorption was calculated from the enclosed area between the loading and unloading curves. The ultimate stress was selected as the maximum stress during loading.

Data Preprocessing: The procedure reported in ref. [22] was used to generate spinodal microstructure data. Detailed information on the data generation process can be found in Supporting Information. Briefly, the procedure allows generation of random microstructures by inputting three phase angles as control parameters. After obtaining the 3D microstructures, represented as 3D arrays $\{\Phi_{ijk}\}^{N,N,N}_{i,j,k=1,1,1}$, where $N = 51$ was the number of grid points along each edge of the cubic microstructure, the dimensionality of the data was reduced by slicing the structures and extracting their cross-sections, denoted as $\{\phi^{(rs)}_{ij}\}^{3,n,N,N}_{r,s,i,j=1,1}$, where r indicates the normal direction, s the index of the cross-section, i and j the indices of the grid points, and $n = 7$ the number of cross-sections in one direction. Representative cross-sections for three microstructures were plotted in Figure S6 (Supporting Information). For simplicity, indices i and j were omitted in other parts of this paper for conciseness.

After generating the microstructures, one sample from each set of phase angles was selected to conduct the tests. The remaining microstructures, which serve to augment the experimental dataset, were matched to the stress–strain relationship according to their phase angles. After obtaining the paired stress–strain data from experiments, the stress values were interpolated at fixed strain intervals (90, 0.006, 0.012, ..., 0.3) for all samples and used them in the DeepONet training and testing.

Two-Step DeepONet (QR) with Uncertainty Quantification: The two-step DeepONet architecture (QR) is presented in Figure S4 (Supporting Information). The branch Net in the model, implemented as a convolutional neural network (CNN), processed the 2D cross-sections of the microstructures. The branch Net input had one channel, taking an individual cross-section as its input and outputting a vector of size m. The $3n$ vectors (each with size m) were fed into the equivariance-preserving unit. They first went through an element-wise max-pooling within the three groups of n vectors. This pooling was designed to achieve invariance to permutations of the cross-sections. After this step, there were three vectors, each with dimension m representing the information in one spatial direction. Then, they were sorted in an element-wise way, resulting in again three vectors as the structural encoding. The directional encoding was one of the three (unsorted) vectors, the choice of which depends on the stress component that was calculated. The four vectors were flattened into one single vector with size $4m$ and finally fed into a shallow fully connected neural network, obtaining the output vector with size p. The trunk net input was the combination of the loading strain ϵ and the pseudo-time t representing the loading/unloading stage. It was fed into a fully connected neural network with output size p. The final output of the entire DeepONet, calculated by the dot product of the two vectors with size p, was the stress prediction $\hat{\sigma}$ corresponding to the input microstructure $\{\phi^{(rs)}\}^{3,n}_{r,s=1,1}$, evaluated at the strain ϵ and loading stage t.

To train the network, the loss function was defined as the mean squared error (MSE) between the experimental and predicted stress–strain responses. To accommodate the data for the neural network, the stress data were normalized to be $\mathcal{O}(1)$, by $\tilde{\sigma} = \sigma/\sigma_0$, where the quantities with tilde refer to the normalized values used for the two-step DeepONet, those without the tilde refer to the original values, and $\sigma_0 = 50$MPa. The learning rate was 10^{-4}. The trunk Net was trained first through 8×10^4 epochs, and then the branch Net was trained through 8×10^4 epochs. More technical

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details of the DeepONet architecture are explained in the Supporting Information.

Uncertainty quantification was examined in this work. Uncertainties arise from two primary sources: The first uncertainty comes from the experimental data, as repeated experiments for the same microstructure yielded non-identical albeit very similar stress–strain curves (see Figure S1, Supporting Information). The second uncertainty originated from the random seed used in ML, which influences the initialization of the learning variables. To account for both types of uncertainties, six surrogate models were constructed. Different random seeds were applied to these six models to capture the ML-related uncertainty. Additionally, since a single microstructure could correspond to multiple experimental stress–strain curves, a random index was introduced to select one stress–strain curve for each microstructure, creating different data pairs across the surrogate models to account for data uncertainty. Once the six surrogate models were trained, they were employed to predict the mean and standard deviation of stresses at different strain values for a given microstructure. By inputting a single microstructure into the six models, six corresponding stress–strain curves were generated. The mean and standard deviation were then calculated from these curves, providing a measure of uncertainty.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Author Contributions

H.-J., B.-Z., Q.-C. and E.-Z., contributed equally to this work. H.J. and E.Z. initiated the collaboration. H.D.E, G.E.K., H.J. and E.Z discussed and conceived the research idea. B.Z. and H.J. fabricated the samples and performed the in situ SEM experiments. E.Z., Q.C. and A.B. developed the ML algorithms. E.Z. developed the quivariance-preserving unit. Q.C. developed the uncertainty quantification. S.K. guided the two-photon printing. All authors analyzed the ML outcomes. H.J. wrote the initial draft with inputs from other authors. H.D.E, G.E.K. and S.K. supervised the work. All authors contributed to analyzing the data, revising the manuscript, and giving final approval for the publication.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that supports the findings of this study are available from the corresponding author upon reasonable request. A PyTorch implementation for DeepONet, Two-step DeepONet (qr), Two-step DeepONet (svd) and FNO is available at: https://github.com/qianyingcao/DeepONetSpinodal Codes for the UNet, ViTO and PCADeepONet are available at: https://github.com/aniruddhabora/Characterization-and-Inverse-Design-of-Stochastic-Mechanical-Metamaterials-UNet-ViTO-PCADON-.

Keywords

inverse design, in situ micro-mechanical experiments, mechanical metamaterial, neural operators, stochastic microstructure

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